

From Web Basics to Digital Makerspaces

ICT Summer School 2026



**Hochschule
Bonn-Rhein-Sieg**
University of Applied Sciences

**{ FACHBEREICH
INFORMATIK }**



EN



André Kless



<< My workspaces

From Web Basics to Digital Makerspaces: Building Personal Web Applications TX00GY05-3001

Search from my workspaces



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Discussion

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PREFERENCES

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Documents

Filename	Last modified	Edited by	
André Kless https://www.h-brs.de/en/inf/andre-kless	10.08.2026 15.26	André Kless	
Day 1 Slides https://docs.google.com/presentation/d/1BNXEhfMibMkQyK7rgUJnINTxJJmDKYzzgT-izlyKto/edit?usp=sharing	10.08.2026 15.18	André Kless	

About Me



André Kless, Lecturer

 41 years old

 Father of four children

 Teaching programming, software engineering, and web engineering

 Passionate about making software development accessible to everyone

Who are you?

Why the Web?

The Web is one of the world's largest software platforms.



WIKIPEDIA
Celebrating 25 years



They all run in a web browser.

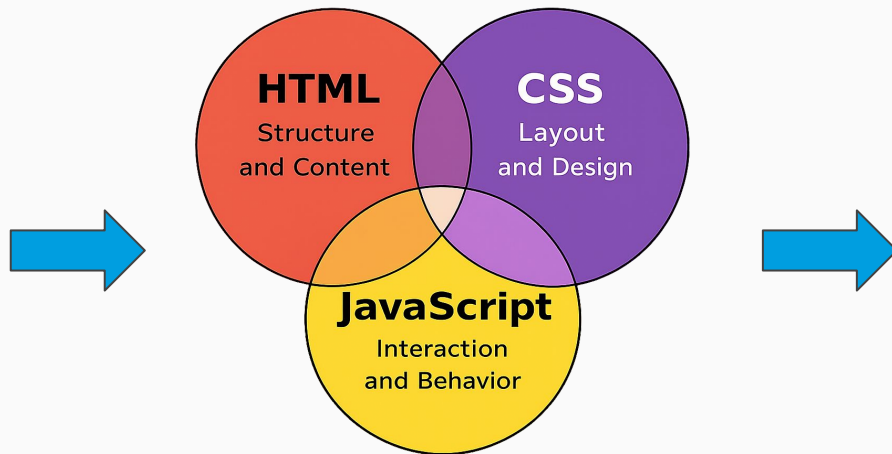
There are many ways to create a website...

Website Builders

- Squarespace
- Webflow
- Wix
- WordPress

AI Assistants

- ChatGPT
- Claude Code
- Gemini
- Lovable
- Bolt



Open Web Standards



HTML

Hypertext Markup Language

HTML – Create the structure

- Defines the structure and meaning of the content
- Organizes content into elements
- Provides the semantic foundation of a web page



CSS

Cascading Style Sheet

CSS – Style the website

- Controls the visual appearance and layout
- Applies styles to web page elements
- Makes web pages responsive to different screen sizes



Javascript

JavaScript – Add interactivity

- Adds behavior and interaction
- Responds to user actions
- Dynamically changes the content and appearance

Course Overview

During this week, you will build your own personal website:

Day	Topic	Outcome
Monday	Web Basics & HTML	Your first website
Tuesday	CSS & Responsive Design	A beautiful website
Wednesday	JavaScript	An interactive website
Thursday	Digital Makerspaces	Embedded web applications
Friday	Data Sovereignty & Web Desktop	Looking beyond today's Web
Online Session 1	Improve your project	Feedback & support
Online Session 2	Final presentation	Showcase your website

Goal: Build and present your own personal website.

Assessment

To successfully complete the course you need to:

- ✓ actively participate
- ✓ build your own personal website
- ✓ present your website during the final online session

Question

Does everyone only need a pass/fail certificate, or does anyone need a graded assessment for their home university or funding programme?



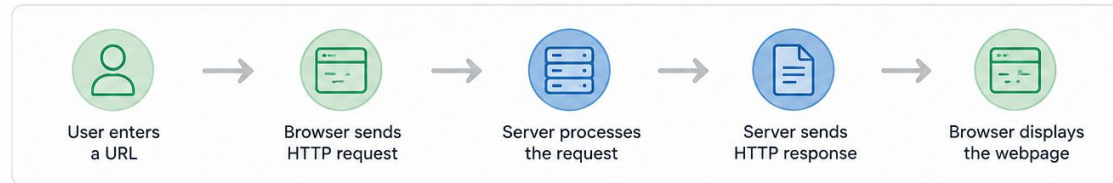
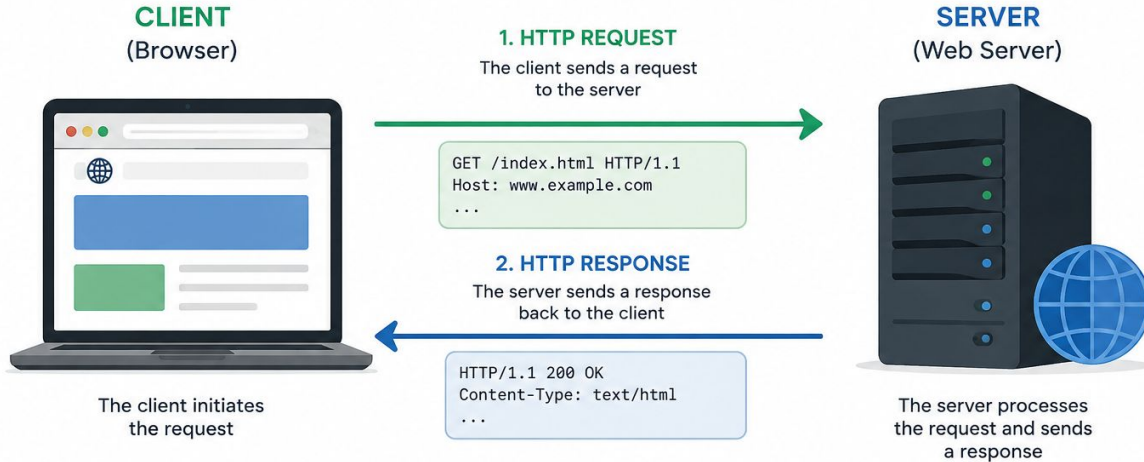
After the course, anyone who wants to can join a board game session.



How does the Web work?

CLIENT-SERVER ARCHITECTURE

Communication with HTTP



The client (web browser) always starts the communication.

The server only responds.

HTTP is the language browsers and servers use to communicate.

When you open a website, the server responds by sending HTML, CSS, and JavaScript to the browser.

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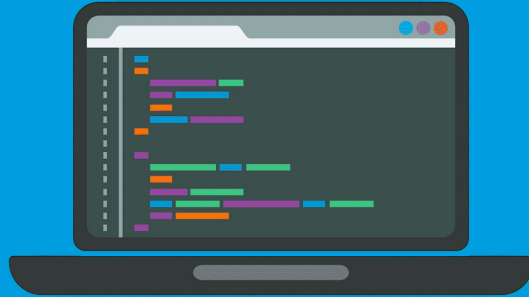


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Live Coding



HTML

Choosing a Code Editor

Editor	Description	Link
Visual Studio Code	Free, cross-platform, beginner-friendly, and extensible.	https://code.visualstudio.com/
WebStorm	Professional IDE for web development (free for students and teachers).	https://www.jetbrains.com/webstorm/
W3Schools Tryit Editor	Browser-based editor – no installation required – perfect for quick experiments.	https://www.w3schools.com/tryit/
Sublime Text	Lightweight, fast, and cross-platform text editor.	https://www.sublimetext.com/
Notepad++	Lightweight text editor for Windows.	https://notepad-plus-plus.org/

Learning Resources

Our primary learning resource:



W3Schools
<https://www.w3schools.com>

Why W3Schools?

- ✓ Beginner-friendly
- ✓ Short explanations
- ✓ Interactive examples
- ✓ Try it yourself in the browser
- ✓ Free to use

Don't be afraid to experiment. That's one of the best ways to learn web development.

Other useful references:



MDN web docs
moz://a

MDN Web Docs
<https://developer.mozilla.org>

Comprehensive reference and documentation for web technologies.

You don't have to remember everything.
You just need to know where to find it.

HTML Recap

What did we learn today?

- ✓ How the Web works
- ✓ How browsers and servers communicate
- ✓ What HTML is
- ✓ How to build your first webpage

Key Takeaways

- HTML describes the structure of a webpage.
- Browsers render HTML into visible content.
- Every website starts with HTML.

The HTML Structure

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>My Personal Website</title>
</head>

<body>

  <!-- Visible content goes here -->

</body>

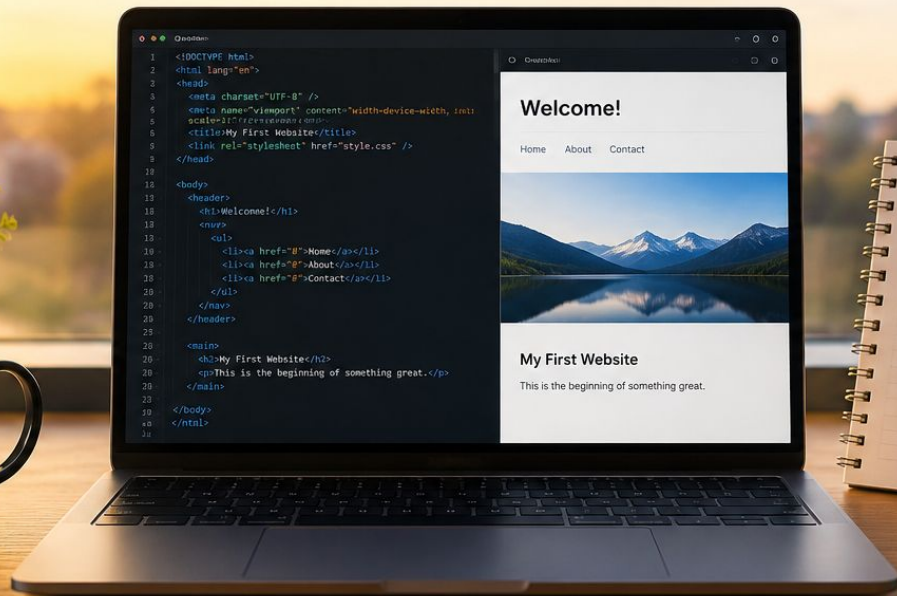
</html>
```

```
<body>
  <header>
    <h1>My Personal Website</h1>
    <nav>
      <a href="...">About Me</a>
      <a href="...">Interests</a>
      <a href="...">Contact</a>
    </nav>
  </header>
  <main>
    <section>
      <h2>About Me</h2>
      <article>
        <h3>My Background</h3>
        <p>...</p>
      </article>
      <article>
        <h3>My Interests</h3>
        <p>...</p>
      </article>
    </section>
    <aside>
      <h2>Did you know?</h2>
      <p>Some additional information...</p>
    </aside>
  </main>
  <footer>
    <p>© 2026 André Kless</p>
  </footer>
</body>
```

Useful HTML tags

<h1> <h2> <h3> <h4> <h5> <h6> <p>
 <a> <audio>
<video> <div> <table> <tr> <td> <th> <form>
<input> <button> <label> <textarea> <select> <option> <fieldset>
<legend> <dialog>

Let's start!



Code.
Create.
Inspire.
♡

Plan ☒
Code ☒
Test ☒
Improve ☒
Launch 

You can
do this!
♡

Have
fun! 😊

Today you created...



Let's make it beautiful!